Week 1 (2/16-2/22, 2026)

Coding

3D visualization of the host/subhalo initial conditions

data: `hosthalo_IC.hdf5`, `subhalo_IC.hdf5`

helpful resources:

3D scatterplot — Matplotlib 3.10.8 documentation

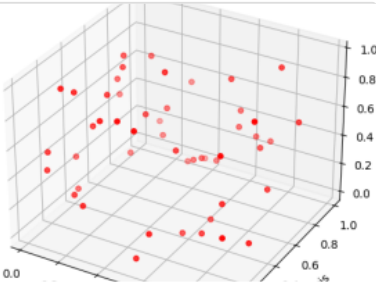
© Copyright 2002–2012 John Hunter, Darren Dale, Eric Firing, Michael Droettboom and the Matplotlib development team; 2012–2025 The Matplotlib development team.

https://matplotlib.org/stable/gallery/mplot3d/scatter3d.html

3D Scatter Plotting in Python using Matplotlib - GeeksforGeeks

Your All-in-One Learning Portal: GeeksforGeeks is a comprehensive educational platform that empowers learners across domains-spanning computer science and programming, school education,

https://www.geeksforgeeks.org/python/3d-scatter-plotting-in-python-using-matplotlib/



wikipedia to help understand what is .hdf5 format:

Hierarchical Data Format

Hierarchical Data Format (HDF) is a set of file formats designed to store and organize large amounts of data. Originally developed at the U.S. National Center for Supercomputing Applications, it is

https://en.wikipedia.org/wiki/Hierarchical_Data_Format

Generate static NFW host halo txt file (for running N-body simulations)

The format of this .txt file is four column, {radius r [kpc], mass density at that radius ρ(r) [M⊙/kpc3], enclosed mass with in the radius M(<r) [M⊙], radial velocity dispersion σv (r) [km/s] }.

helpful resources:

numpy.linspace — NumPy v2.5.dev0 Manual

Returns num evenly spaced samples, calculated over the interval [start, stop].

https://numpy.org/devdocs/reference/generated/numpy.linspace.html

numpy.column_stack — NumPy v2.4 Manual

Take a sequence of 1-D arrays and stack them as columns to make a single 2-D array. 2-D arrays are stacked as-is, just like with hstack. 1-D arrays are turned into 2-D columns

https://numpy.org/doc/stable/reference/generated/numpy.column_stack.html

numpy.savetxt — NumPy v2.1 Manual

If the filename ends in .gz, the file is automatically saved in compressed gzip format. loadtxt understands gzipped files transparently.

https://numpy.org/doc/2.1/reference/generated/numpy.savetxt.html

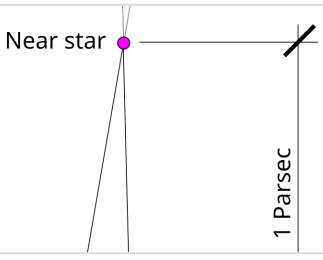
Reading

(Understanding the unit: what is a “parsec”?)

Parsec

The parsec is a unit of length used to measure the large distances to astronomical objects outside the Solar System, approximately equal to 3.26 light-years or 206265 astronomical

https://en.wikipedia.org/wiki/Parsec



(SIDM wiki:)

Self-interacting dark matter

In astrophysics and particle physics, self-interacting dark matter (SIDM) is an alternative class of dark matter particles that have strong interactions, in contrast to the standard cold dark matter model (CDM). SIDM was postulated in 2000[1] as a solution to the core-cusp

https://en.wikipedia.org/wiki/Self-interacting_dark_matter

(Core-cusp problem:)

Cuspy halo problem

The cuspy halo problem (also known as the core-cusp problem) is a discrepancy between the inferred dark matter density profiles of low-mass galaxies and the density profiles predicted by cosmological N-body simulations. Nearly all simulations form dark matter halos which have "cuspy" dark matter distributions,

https://en.wikipedia.org/wiki/Cuspy_halo_problem

- **read sections 1.1-1.2, 3.2.2 (1.3-1.4)**

Small-Scale Challenges to the $Λ$CDM Paradigm

The dark energy plus cold dark matter ($Λ$CDM) cosmological model has been a demonstrably successful framework for predicting and explaining the large-scale structure of Universe

https://arxiv.org/abs/1707.04256



- **read section I (III)**

Dark Matter Self-interactions and Small Scale Structure

We review theories of dark matter (DM) beyond the collisionless paradigm, known as self-interacting dark matter (SIDM), and their observable implications for astrophysical structure in the...

https://arxiv.org/abs/1705.02358



- **(read III A&B)**

Astrophysical Tests of Dark Matter Self-Interactions

Self-interacting dark matter (SIDM) arises generically in scenarios for physics beyond the Standard Model that have dark sectors with light mediators or strong dynamics. The self-interactions...

https://arxiv.org/abs/2207.10638



(write notes: What is the standard dark matter picture/model? What problems does our standard DM model face at small scales? What is "small scale"? How can SIDM potentially solve the problems? What is SIDM? What parameters do we use/discuss/study for the SIDM model?)